Software Requirements Specification: AI-Powered Hospital Management System (HMS)

1. Project Overview

The AI-Powered Hospital Management System is a centralized digital platform designed to streamline administrative, clinical, and operational workflows for hospitals and clinics. It integrates AI-driven insights to assist in patient triage, resource optimization, and clinical decision support.

2. Executive Summary

The system aims to reduce administrative burden on medical staff, improve patient flow, and enhance diagnostic accuracy through predictive analytics. Key capabilities include digitized medical records, scheduling, AI-assisted diagnosis support, and automated resource allocation.

3. Stakeholders

- **Hospital Administrators:** Oversee operational efficiency and resource management.
- **Physicians/Clinicians:** Use the system for patient care, diagnosis, and prescription.
- **Nursing Staff:** Manage patient vitals, medication administration, and record keeping.
- **Patients:** Interact with the system for scheduling and record access.
- **IT/Support Staff:** Manage system maintenance and security.

4. Target Users

- **Admin/Reception:** Manage appointments, billing, and registrations.
- **Medical Professionals:** Access electronic health records (EHR) and AI-driven clinical insights.
- **Patients:** View their own medical history and appointment status.

5. Functional Requirements

- **FR1: Patient Management:** Registration, intake, and centralized medical history.
- **FR2: Appointment Scheduling:** Automated booking, rescheduling, and notification system.
- **FR3: AI-Assisted Triage:** Use AI to categorize patient urgency based on symptoms.

- **FR4: Clinical Decision Support:** AI-powered suggestions for diagnostic patterns based on patient history.
- **FR5: Electronic Health Records (EHR):** Secure storage and retrieval of test results, treatment history, and prescriptions.
- **FR6: Resource Management:** AI-based predictive modeling for bed occupancy and staffing needs.
- **FR7: Billing & Invoicing:** Generation of invoices and tracking of insurance claims.

6. Non-Functional Requirements

- **NFR1: Security:** End-to-end encryption for all sensitive health data (HIPAA compliance).
- **NFR2: Reliability:** 99.9% uptime requirement for critical clinical functions.
- **NFR3: Scalability:** Ability to handle concurrent users across multiple clinic branches.
- **NFR4: Performance:** System response time for clinical data retrieval under 2 seconds.
- **NFR5: Usability:** Intuitive UI optimized for high-pressure medical environments.

7. User Stories

- **As a Receptionist**, I want to book appointments efficiently so that I can reduce patient wait times.
- **As a Doctor**, I want an AI-summary of a patient’s historical records before a consultation to save time.
- **As a Patient**, I want to receive SMS notifications for my upcoming visits.
- **As an Administrator**, I want to see a daily dashboard of bed availability to optimize staffing.

8. Acceptance Criteria

- The system must correctly identify patient priority levels with at least 90% accuracy according to established triage protocols.
- System updates to medical records must reflect in real-time across all user dashboards.
- All user actions must be recorded in an immutable audit log.

9. Business Rules

- Access to medical records is role-based (e.g., Doctors can edit, Receptionists can only view demographic data).

- All AI suggestions must include a "Human-in-the-loop" verification button that clinicians must sign off on.
- Patients must provide explicit consent for data usage in diagnostic training.

10. Assumptions

- The hospital has stable internet connectivity.
- All medical staff possess basic digital literacy.
- Regional laws permit the use of AI for clinical support provided human oversight exists.

11. Constraints

- Must comply with local data privacy regulations (e.g., HIPAA, GDPR).
- Integration with existing legacy hardware (e.g., imaging machines) may be required.

12. Risks

- **Data Privacy Breach:** Misuse or unauthorized access to sensitive PHI (Protected Health Information).
- **AI Hallucination:** Incorrect diagnostic suggestions leading to clinical error.
- **User Resistance:** Medical staff may find AI tools intrusive to their workflow.

13. Out of Scope

- Direct hardware manufacturing (e.g., sensors or MRI machines).
- Development of a native patient mobile app (Web-only for the initial phase).
- Pharmaceutical supply chain management.

14. Open Questions

- What legacy systems (if any) need to be integrated?
- Which specific medical fields (e.g., Oncology, Cardiology) should be prioritized for AI training?
- What is the specific regional regulatory standard for data storage duration?